

ENGINEERING REBOOT 2026

Week 06 – Day 27 Field Manual

Publication Manuscript v1.0

Status: LOCKED

Authority: Week 06 Baseline

1. Mission

Today's objective is to strengthen practical Bash proficiency by learning how commands, variables, redirection, permissions, and the shell environment work together to automate common engineering tasks. Building on Day 26, today's work transitions from operating within Linux to using Bash as an engineering automation tool.

2. Engineering Context

Modern software engineers rely heavily on command-line environments for Linux servers, Docker, CI/CD, cloud infrastructure, Git, build systems, and Python environments. Bash enables safe, repeatable automation of these workflows.

3. Learning Objectives

Technical: variables, command substitution, quoting, pipes, redirection, permissions, environment variables, aliases, reusable scripts. Engineering: automate repetitive work, write maintainable and defensive scripts.

4. Core Concepts

Variables, positional parameters, command substitution, exit status, PATH, quoting, stdin/stdout/stderr, pipes, redirection, chmod, chown (conceptual), aliases, executable scripts, shebang.

5. Exercises

Implement exercises covering variables, command substitution, quoting, pipes, redirection, permissions, environment variables, and aliases. Detailed scenarios will be addressed during implementation.

6. Mini Project

Build a Bash System Information Reporter that gathers workstation information, displays it cleanly, and writes a timestamped report using variables, command substitution, conditionals, and output redirection.

7. Validation Matrix

Validate variable expansion, command substitution, pipes, report generation, execution permissions, successful repeated execution, and correctness of generated system information.

8. Evidence Checklist

Capture exercise output, source code, execution results, generated report, validation evidence, Notes.md, Journal.md, Git commit/push/fetch, and repository verification.

9. Engineering Review

Assess functionality, readability, organization, naming consistency, maintainability, defensive scripting, and appropriate Bash idioms.

10. Notes.md Guidance

Record observations on variables, command substitution, quoting, pipes, redirection, permissions, environment variables, Bash practices, and differences between sh and Bash.

11. Journal.md Guidance

Reflect on familiar Bourne shell concepts, new Bash capabilities, useful automation techniques, and areas requiring further practice.

12. Completion Checklist

Exercises, mini-project, validation, evidence, Notes.md, Journal.md, Git operations, and repository verification completed.

13. Preview

Day 28 extends Bash scripting into broader Linux engineering workflows involving process management, package management, and system administration concepts.